

Please create issues during the procurement process to get it in our queue for scheduling purposes.

The (future) tech stack application business owner or technical owner is responsible for creating the change management issue and providing approvals and technical review during the process.

- Procurement Discovery and PoC

- Verify Okta SAML/SCIM feature support: with vendor technical contact or in the product documentation.

- Complex Applications: For complex or mission critical applications, it is recommended to start an email or Slack thread to discuss in collaboration with the Corporate Security Identity team. It is always better to engage us early instead of waiting until a few weeks before you want to go live.

- PoC or Sandbox Instance: If you have a pre-procurement environment/instance, you can create a change management issue for a test application instance to be added. We do not have rushed timelines available, so please do not treat this as a blocker for your PoC to be completed.

- Procurement Negotiations

- Create the change management issue to get it the first in, first out queue. We cannot begin implementation or go live for users until the procurement process is completed.

- Add details about the expected Okta groups and users that should be assigned to the application.

- For complex applications, add details about roles and permissions for users. This impacts how we assign and/or create Okta groups for each role. This is usually discussed in depth during the discovery call.

- Vendor Onboarding

- Add the new application to the tech stack

- Update the change management issue with the tech stack details and comment that the procurement process is complete.

- The Corporate Security Identity Team will schedule a technical discovery call if it hasn't happened already.

- Application Creation

- See the change management issue for the process. This process can take 1-2 weeks with approvals and testing.

- The go live process is flexible and varies based on the application. Please comment in the change management issue with expectations and also discuss on the discovery call.

Configuration Updates

All configuration updates are managed using a change management issue.

Update Application Change Management Issue Template

Access Requests

All group and user updates are managed with access requests.

- Okta Access Requests

- Okta Groups

SECTION: security/corporate/systems/okta/ar/_index.md

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SECTION: security/corporate/systems/okta/group/_index.md

This is a placeholder page. Please see the links below for any child pages that exist.

SECTION: security/corporate/systems/okta/group/members/_index.md

Overview

Managers can self service add users to Google Groups that are synced to Okta for access to various Okta applications.

If you are listed as an Owner/Provisioner for an application in the tech stack you will be using the method below to add users to a Google group, which will then sync this group to Okta and assign the application to users. This process was created to empower business application owners to effect Access Requests which require Okta application assignment.

- Sign in to Google Groups.
- Click My groups.
- Click the name of the group were you want to add/remove a user. (Note that all Google groups which manage users in Okta application start with okta-xxxxx-users)
- Next press the People tab on the left side and select Members.

- This will show you all the members currently in the group.
- To add a member press the Add Members button. To remove access mouse over a user and press on the little white box that appears, this will mark the user. After that on the right side press the remove member button (Looks like a circle with a horizontal line across).

When a member is added/removed from the group it may take up to 1 hour for the sync to happen between Google and Okta. Once the sync happens the user will see the application in Okta, if removed the opposite.

If you have any questions or require assistance please reach out to the IT team in the it-help Slack channel.

SECTION: security/corporate/systems/okta/group/nomenclature/_index.md

Overview

Okta Groups need to follow standardized naming conventions for better discovery, tracking, and automation potential.

Syntax for user-metadata groups

{departmentslug}.{functionalteamslug}.{specificroleifapplicable}

Example: sales.accountexec.amer

Syntax punctuation

- periods (.) as separators
- underscores () for spaces
- not using hyphens (-) or camelCase

App assignments groups

app.{system}.{environment}.{role/permission}

Example: app.salesforce.prod.accountexecemea

App approver group

corpsys.approver.{system}

Example: corpsys.approver.salesforce

App provisioner group

corpsys.provisioner.{system}

Example: corpsys.provisioner.salesforce

SECTION: security/corporate/systems/okta/group/rules/_index.md

This is a placeholder page. Please see the links below for any child pages that exist.

SECTION: security/corporate/systems/okta/group/rules/nomenclature/_index.md

Overview

Okta Groups Rules need to follow standardized naming conventions for better discovery, management, and auditing.

Guide

When appropriate, a group rule name should be the same as the group it manages membership to.

> See Okta Group Nomenclature for more information about naming syntax.

Example:

- Group Rule: app.googleworkspace.employees
- conditions: user.userType=="Employee"
- groups: app.googleworkspace.employees

When a group rule manages membership for multiple groups, the group rule name should descriptively and succinctly describe the group rule's purpose.

Example:

- Group Rules: Corporate Security Department
- conditions: user.department=="Corporate Security"
- groups: dept.corporatesecurity,div.security,app.gitlab.corporatesecurity

SECTION: security/corporate/systems/okta/org/_index.md

SECTION: security/corporate/systems/okta/user/_index.md

This is a placeholder page. Please see the links below for any child pages that exist.

SECTION: security/corporate/systems/okta/user/android/_index.md

<!-- FIXME -->

Google Workspace 2FA Prompts Missing

> When adding my GitLab Google Workspace account to my Android device, the biometrics or security key verification never prompts after signing in to Okta.

When attempting to add a Google Workspace account to an Android device, Okta authentication proceeds in the Android's embedded browser (WebView).

Since Okta does not support embedded web browsers for WebAuthn based verification, which causes an issue where nothing prompts you after you sign-in to Okta, so you cannot add the Google Workspace account to Android devices.

Please reach out to ithelp Slack channel to request temporary enable Okta Verify as a

workaround.

SECTION: security/corporate/systems/okta/user/faq/_index.md

Frequently Asked Questions

If you have a question that is not answered here, please ask in ithelp on Slack or create an issue in the Corporate Security Issue Tracker.

GitLab.com 2FA

> Why does GitLab.com ask for an additional MFA when I login using Okta?

Your gitlab.com account will have 2FA installed as required by our policy.

Note that the 2FA for GitLab.com is different to the MFA you use to log into Okta.

This issue has been opened to propose a solution.

Android Authentication Errors

> When adding my GitLab Google Workspace account to my Android device, the biometrics or security key verification never prompts after signing in to Okta.

When attempting to add a Google Workspace account to an Android device, Okta authentication proceeds in the Android's embedded browser (WebView).

Since Okta does not support embedded web browsers for WebAuthn based verification, which causes an issue where nothing prompts you after you sign-in to Okta, so you cannot add the Google Workspace account to Android devices.

Please reach out to the ithelp Slack channel to request us to temporarily enable Okta Verify as a workaround.

Browser Plugin

The Okta Browser Plugin is optional.

GitLab is using Secure Web Authentication (SWA) for some of its apps, and the Okta plugin is necessary to work with these applications. For applications that support SAML (most of them), the plugin is not necessary.

Vendor Docs

How can I change the order in which my apps appear?

You can click and hold on an app icon to drag and drop the app to the location that you would like it to be displayed.

Bookmarks

A bookmark is a way to save the URL login of an app not currently available to you. When you create a new bookmark, your Okta dashboard will display an app icon to that app URL login. This bookmark will only store the URL, not your username and password.

To add a bookmark, go to the +Add Apps button on the top right of your dashboard to open a search menu. Search for your app. If no app is found, Okta will display the option to create a bookmark. Enter the URL of the app and the name of the bookmark you would like displayed. Click the Add button to create your bookmark. Click on Home at the top of your dashboard to see your new bookmark.

Remove Unused App

> How do I remove an app?

Apps that you've installed can be removed by hovering over the app and selecting the gear icon. In the app setting screen, you'll see the option to delete it.

Apps granted by Corporate Security or the tech stack owner cannot be removed, but there is a way to move the app out of sight. The best way to do so is to create a new tab to store unused and unwanted applications.

Searching for Apps

> How do I search an app if I can't remember which tab I put it on?

At the top of your dashboard, you can find apps in the Launch App search bar. If you can't remember which tab your app is on, go to the Launch App search, type in the name of your app and select the name to open it when it appears.

Application Data Privacy

> Does Okta store our information entered into apps like Workday, NexTravel, Carta etc?

No, Okta only acts as an integrator to the various apps. The information that the user stores inside of the app, is not accessible by Okta. The usernames and passwords (user credentials) are encrypted using both an industry--standard encrypted AES and a randomly generated symmetric key.

Session Expiration

> How long does my login last until I need to re-authenticate?

Okta does not log you out of your applications even though you might be logged out of your Okta session.

Okta has two controlled re-authentication parameters -- Factor Lifetime and Session Lifetime.

Factor Lifetime: Setting a factor lifetime is a way for end users to sign out for the amount of

time noted in the Factor Lifetime and not have to authenticate again with MFA at the next sign in. End users must check a box to confirm that the setting should be applied. We have configured this to be 15 minutes. If you are logging on with a machine that isn't your usual device, make sure you don't check this button!

Session Lifetime: The maximum idle time before an authentication prompt is triggered. We have configured this to be 16 hrs. This is the more significant factor to note in relation to authentication, and should mean that if you log into the Okta dashboard using MFA, you should not be prompted for any additional authentication during your working day unless a specific application requires you to re-auth. For most people, this creates a straightforward daily process where they log into Okta once at the start of their day, and can keep their browser open to access applications for the rest of the day.

If you are having issues with excessive authentication requests, please log an issue and we will investigate.

Google OAUTH

> Is it acceptable to continue using Google Oauth login for some of the apps (with its own 2FA)?

Yes, this is fine. Our end state will be to use Okta for most/all applications, but in the meantime existing authentication methods are fine. Once we have broader user adoption of Okta, we'll be taking on migration of some of these apps to use Okta instead of Google OAuth, where this makes the most sense.

Outages

When Okta is down, basically GitLab will be down. Any mitigation strategy would enlarge our security surface area.

Okta is built on an always on architecture. However, if their services were to go down, you would not be able to log in to Okta or access your applications using single sign-on.

Many applications have long lived sessions (ex. Slack, Gmail, GitLab) so it is likely that although not all applications would work, you could still get by with a few, or take a break for a while.

Any outages or incidents that are significantly disruptive will be posted in whats-happening-at-gitlab on Slack.

Emergency Contacts

We have 24x5 coverage in ithelp on Slack. Our Helpdesk Analysts will escalate to the CorpSec Engineers as needed.

In a Slack thread, tag @sysowners-okta. In a GitLab issue, tag @gitlab-com/gl-security/corp/systems/okta. You can hover over these group names in the respective app to see the latest list of members.

Weekend response is based on a best effort and will likely have a delayed response.

SECTION: security/corporate/systems/okta/user/ios/_index.md

<!-- FIXME -->

These steps are for an iPhone, and may be slightly different for Android. If you are using an iPhone and receive a Developer or XCODE error, please upgrade to iOS 16+. We recommend enrolling a phone even if you don't plan to use it often, in case you need a way to add a new computer or your credential gets accidentally removed on the computer.

1. On the computer, if using Chrome, please check that it is on the latest version by visiting the URL <chrome://settings/help> - if a new version is available, please use the Relaunch button to restart the browser.
2. On your Mac, please also check that you are using macOS Ventura 13.0.1 or newer.
3. On your Mac, please check under "System Settings"->"Privacy & Security"->"Bluetooth" and make sure that Google Chrome has Bluetooth access enabled.

4. On the computer, login into your Okta
5. On the computer, click on your name on the top right to open the drop down menu (similar to above) and navigate to "Settings".
6. On the computer, under Security Methods click "Set up another" beside Security Key or Biometric. This will take you to a setup authentication screen, click "Verify". Provide Touch ID.
7. On the computer, click "Set up", then click "Set up" again on the next screen
8. On the computer, ignore all popups from the 1Password browser extension as it could interfere with the process of transferring the passkey. Click the "Yubikey" icon to bypass if it appears.

1. On the computer, the next step depends on the browser and version.
 1. In Chrome, the pop up that opens states "Create a passkey for gitlab.okta.com" and has your email address listed. Choose "Use a different passkey" at the bottom left.

<img src="/images/business-technology/okta/Okta-Chrome-Passkey-Popup.png" alt="Okta Chrome

Passkey" width="300"/>

Then, choose "Use a phone, tablet or security key".

This should display a QR code that you can scan.

2. In Safari, click "Other options", then choose "iPhone, iPad or Android device - save a passkey on a device with a camera", then "Continue".

2. On the mobile device, open your camera app and scan. (It may take a few seconds for it to connect). This requires Bluetooth to be enabled on both devices, but does not require pairing.

3. On the mobile device, a pop should show up to allow the credential to be saved.

1. On a iPhone or iPad, it will prompt to save the credential to iCloud Key Chain. Allow this.

2. If saving the credential fails, go to Settings->your name->iCloud and make sure that Passwords and Keychain is set to "On".

3. If it still fails, you can also try force restart and then trying to sign out of iCloud on your mobile device, and then sign back in to iCloud.

4. On Android, this may appear "Use this device with screen lock". Choose this.

4. On the mobile device, attempt to sign in by visiting Okta.

SECTION: security/corporate/systems/okta/user/linux/_index.md

This is a placeholder page. Please see the links below for any child pages that exist.

SECTION: security/corporate/systems/okta/user/lockout/_index.md

Instructions

1. Ask for a Okta unlock (or 2FA reset) in the ithelp Slack channel or send an email to it-help@gitlab.com and ask for a 2FA Reset.

> If you do not have access to your work email account, you can send from your personal email address that is on file in Workday.

2. Please be prepared to verify your identity.

3. Once Okta 2FA has been reset, please reconfigure it by signing in to your Okta account and adding a new authenticator using Touch ID or YubiKey.

SECTION: security/corporate/systems/okta/user/macOS/_index.md

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This method has been verified on Macs and Linux with Chrome. For Safari, it requires macOS Ventura 13+. Steps below for iPhone require iOS 16+, may be slightly different for Android.

1. On the computer, log in with username and password
1. On the computer, a popup appears to "Verify your identity with gitlab.okta.com"
1. On the computer, choose "Use phone with a QR code". This requires Bluetooth to be enabled on both the phone and the laptop, but doesn't require pairing.
1. On the mobile device, scan the code using the Camera app
1. On the mobile device, click "Sign in with a Passkey"
1. On the mobile device, a "Sign in" popup appears - "Do you want to sign in to "gitlab.okta.com" with your saved passkey for "xxxxx@gitlab.com"? Click Continue and provide biometric.
1. On the computer, you will now be signed in to Okta.
1. If applicable, follow the standard steps to enroll your Touch ID into Okta.

I don't have an enrolled phone or computer but have a YubiKey

If both of previous devices are not available, you could use a YubiKey as another form of authentication (if you have one set one up). Use that to access your settings page and follow the steps above to enroll a new device.

SECTION: security/corporate/systems/okta/user/passkey/_index.md

<!-- FIXME -->

1. With the 1Password browser extension installed, access the Settings page.
1. In the 'Security Methods' section of the page, choose Set up or Set up another next to Security Key or Biometric.
1. You may then be presented with another prompt to confirm if you wish to Set up another, followed by an Enroll prompt.
1. After pressing Enroll, a prompt from your web browser will appear.
1. From 1Password, you'll see a pop up notification from 1Password with the options "New Item" or "Update Existing". Select "Update Existing" and your Okta account to continue.

SECTION: security/corporate/systems/okta/user/provisioning/_index.md

This is a placeholder page. Please see the links below for any child pages that exist.

SECTION: security/corporate/systems/okta/user/setup/_index.md

All GitLab team members have an Okta account created as part of the onboarding process.

New Team Member? You should already have an activation email in your personal email inbox.

Authentication

You can access the Okta dashboard by opening your web browser and visiting <https://gitlab.okta.com>. You should bookmark this page. ·

If you visit a vendor's website and choose to sign in with SSO, it will redirect you to Okta to sign in. If prompted for a company ID or similar, try gitlab.

Username

Your GitLab email address is your username. This is usually {firstName}{lastName}@gitlab.com (ex. Dade Murphy would be dmurphy@gitlab.com).

Keep in mind that your GitLab.com (the product) username will probably be different and will not include @gitlab.com. Whenever we refer to your GitLab username, it refers to your product username that can be mentioned in issues and merge requests. Whenever we refer to your email handle, it refers to the first part of your email address before the @.

Password

When you first create your Okta account, you will not have access to 1Password yet to generate a secure password. Please use a memorable long phrase password temporarily.

After your 1Password account has been created during onboarding, you should generate a 64-character random password in 1Password and change your Okta password so that it autofills from 1Password and is not a shorter, memorable password. The only password you should be able to remember is your 1Password Master Password.

After your onboarding tasks are completed, you should not be able to type or memorize your Okta password and it should be autofilled from 1Password.

- 1Password Vendor Docs - Generate Strong Passwords
- Okta Vendor Docs - Change Your Okta Password
- GitLab's Password Policy

2FA

GitLab requires all team members to use either Apple Touch ID/Face ID biometrics or YubiKey that uses the WebAuthn protocol.

For your first few days, don't worry about setting up mobile devices.

For most users, this is simply adding your TouchID fingerprint to Okta for 2FA.

- macOS
 - · macOS Touch ID Guide
 - Okta Verify Device Trust for macOS
 - YubiKey Guide

- Linux
 - Exception Group <!-- TODO -->
 - YubiKey Guide
- Android 2FA
 - Android 2FA Guide
 - Okta Verify Device Trust for Android
- Apple iOS 2FA
 - iOS Face ID 2FA Guide
 - Okta Verify Device Trust for iOS

Access Applications

You can access the Okta dashboard by opening your web browser and visiting <https://gitlab.okta.com>.

The Okta End-User Dashboard allows you to easily access your enterprise applications securely using Single Sign On (SSO) technology.

Vendor Docs - Dashboard Overview

Vendor Docs - Working with Apps

You can click on any of the application tiles that you see on your dashboard to access those applications. For most applications, you will not need to enter a password. Okta automatically passes on access through a token based on a pre-established trust relationship and user provisioning automation.

Onboarding Guide - Collaboration Apps

Onboarding Guide - Employment Apps

For security reasons, Corporate Security has limited the number of applications that are available to access from your Android or iOS mobile device.

Phone Services

Access Requests

You will be granted to additional applications throughout the onboarding process.

Okta Access Requests

SECTION: security/corporate/systems/okta/user/touchid/_index.md

<!-- FIXME -->

Overview

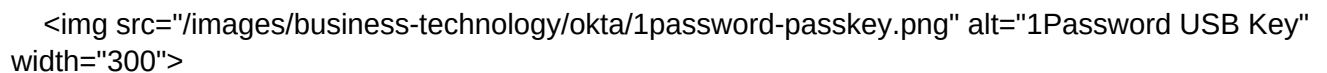
Configuration Steps

1. While logged in to Okta from the device you wish to add, access the Settings page.

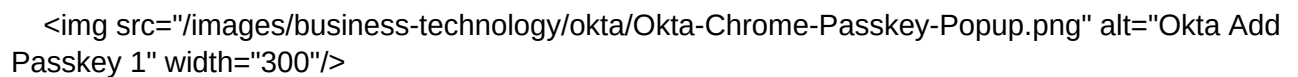
1. In the 'Security Methods' section of the page, choose Set up or Set up another next to Security Key or Biometric.

1. You will then be presented with a password verification and passkey verification prompt.

1. After pressing Set up twice, a prompt from your web browser will appear, or a 1Password prompt may appear. Click the thumb drive icon to bypass if it appears and you wish to save it in iCloud Keychain (recommended) or the local Chrome browser.

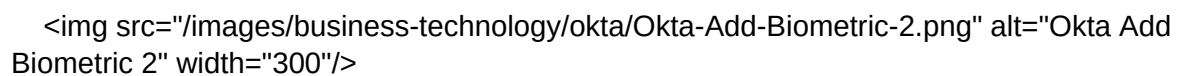
A screenshot of a 1Password USB Key prompt, showing a USB drive icon and text indicating the option to save the passkey in iCloud Keychain or the local Chrome browser.

1. For iCloud Keychain or a YubiKey, click "Use a different passkey". For the local Chrome browser choose "Continue".

A screenshot of the 'Okta Add Passkey 1' dialog box, which prompts the user to choose between 'Use a different passkey' and 'Continue'.

<!-- FIXME -->

1. If you choose to save it in your Chrome profile, another prompt will appear asking you to authenticate using Touch ID or Face ID.

A screenshot of the 'Okta Add Biometric 2' dialog box, which prompts the user to authenticate using Touch ID or Face ID.

1. If you chose to use the iCloud Keychain or Security Key, choose the correct option to finish enrollment.

A screenshot of the 'Okta Add Passkey 2' dialog box, which prompts the user to choose between 'Use a different passkey' and 'Continue'.

1. For Security Key, relevant prompts will appear.

A screenshot of the 'Okta Security Key 1' dialog box, which prompts the user to enter a PIN or use a biometric method to authenticate.

1. You may be prompted for a PIN, and then finally click Allow.

A screenshot of the 'Okta Security Key 2' dialog box, which prompts the user to click 'Allow' to complete the enrollment.

1. We recommend enrolling both Chrome and Safari for redundancy on your computer, as well as a mobile device. Each browser needs to be enrolled separately, unless you use iCloud Keychain. Note that separate Chrome browser profiles also need to be enrolled separately. You should enroll one computer browser (for example, Chrome), then add the mobile device, then if required add the other computer browser (for example, Safari).

1. If clearing your browser cache, please be careful as you can delete your passkey.

1. In Chrome, if using "Clear Browsing Data" function please ensure that you do not check "Passwords and other sign-in data" (under "Advanced" tab). By default, this is unchecked.

2. In Safari, the credential is stored under "Settings->Password". This is separate from the "Settings->Privacy" and "Manage Website Data" where you would click "Remove all" to remove the cache.

We have also prepared Introductory Videos on Configuring Applications and Dashboard Tips.

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SECTION: security/corporate/systems/okta/user/yubikey/_index.md

This is a placeholder page. Please see the links below for any child pages that exist.

SECTION: security/corporate/systems/okta/verify/_index.md

Okta Verify is a multifactor authentication (MFA) app that allows users to confirm their identity when they sign in to their Okta account or access Okta-protected resources.

This is a placeholder page. Please see the links below for any child pages that exist.

SECTION: security/corporate/systems/okta/verify/android/_index.md

This is a placeholder page. Please see the links below for any child pages that exist.

SECTION: security/corporate/systems/okta/verify/ios/_index.md

This is a placeholder page. Please see the links below for any child pages that exist.

SECTION: security/corporate/systems/okta/verify/macOS/_index.md

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Vendor Docs - macOS Setup

Vendor Docs - FAQ and Troubleshooting

Overview

Note: Occasionally after a macOS version update, Okta Verify will need to be set up again. Following these steps will resolve the issue.

1. Open the Okta Verify application on macOS via Spotlight by selecting the magnifying glass in the menu bar (top right corner of display). Another method to open Spotlight is to use cmd + Space

1. Search for Okta Verify

1. Select Get Started

1. Select Next

1. Confirm gitlab.okta.com is populated for Sign-in URL and then select Next

1. A browser window will open to gitlab.okta.com to confirm your identity

1. After doing so, you will see the Your Identity is Verified message and you can close the browser

1. The Okta Verify application will refresh. Press Next if you are using Touch ID on your Mac (most common), or Skip if you are not. Okta Verify uses macOS's Touch ID capability, and the mathematical calculation is stored locally within Apple's Secure Enclave and not available to Okta or GitLab.

1. If you pressed Next, then click Enable Touch ID

1. Your GitLab Okta account will now be available to use with Okta Verify on macOS

After enrolling

You will continue to be able to use a YubiKey in addition to Touch ID to login to Okta from macOS device. You will not be able to login from a personal macOS device.

A macOS device missing the certificate (even when enrolled with Okta Verify), will quickly show:

Then re-direct to the following error:

SECTION: security/corporate/systems/okta/workflows/_index.md

This is a placeholder page. Please see the links below for any child pages that exist.

SECTION: security/corporate/systems/okta/workflows/flows/_index.md

This is a placeholder page. Please see the links below for any child pages that exist.

SECTION: security/corporate/systems/okta/workflows/flows/google-deprovisioner-flow.md

Overview

Google Workspace Deprovisioner consists of a collection of flows built in Okta Workflows ("OWF"). The flows were created to provide a consistent and automated process for deprovisioning Google Workspace user accounts.

The deprovisioner is composed of 2 distinct parts.

Initial Offboarding

The following actions are performed on a Google Workspace user when an offboarding is triggered via a Slack slash command.

- Signs a user out of all web and device sessions and reset their sign-in cookies.
- Update user profile
 - reset password
 - remove from Global Address list
 - remove recovery email
 - remove recovery phone
 - move to /Former Team Members organizational unit
- Remotely wipe corporate Google Workspace data from mobile devices
- Delete access tokens (OAuth tokens)
- Delete application specific passwords (ASPs)
- Delete verification tokens
- Remove user membership from Google groups.
- Create and set a vacation auto-reply in the user's Gmail
- Make manager a Gmail delegate
- Create a transfer request and transfer the following data to the user's manager
 - Google Drive
 - Google Calendar

Post-90 Day Offboarding

The following actions are performed after 90-days for an already offboarded user in Google

Workspace

- Update user profile
 - move to /No Google License organization unit
- Delete all user aliases
- Suspend the user

How to Use

Initial Offboarding

To trigger an initial offboarding use the following Slack slash command associated with the Google Deprovisioner Slack app.

`/google-offboard <user email>`

Note: The Slack slash command is only available to limited members of the Corporate Security team.

{: .note}

A successful trigger will result in a one moment...processing request for <user email> slack message before the offboarding initiates.

A slack thread will be started by the Google Deprovisioner app which will contain all actions performed as part of offboarding. A status and message will be returned for the results of each action.

Once complete Google Deprovisioner will update the thread and report whether or not the offboarding was successful.

Post-90 Day Offboarding

Post 90-day offboarding are automatically triggered after 90 days. Google Deprovisioner will run once a day to check the status of all past offboards and determine if a post-90 day offboarding needs to be run. Reminders will be sent to the user's manager on the 60, 83, and 90 day mark.

Architecture

The Google Workspace deprovisioner is composed of the following components

{{% alert title="Note" color="primary" %}}

The following documentation assumes some experience and working knowledge with Okta Workflows.

{{% /alert %}}

Flows

The Google Workspace deprovisioner is composed of different types of flows. They are defined by the Type at the beginning of each flow's name. For example Orchestrator in Orchestrator - Google Workspace - Offboarding.

- Trigger
 - Flows that trigger the Google Workspace Deprovisioner. Trigger flows may include scheduled runs, slack slash commands, delegate flows, or any other flow that is initiated by a user or system. These flows should trigger the orchestrator flows to start.
- Orchestrator
 - Flows that orchestrates all the various action and events as part of Google Workspace deprovisioning.
 - An orchestrator flow will usually be composed of multiple actions, notifications, and logging flows.
- Action
 - Flows that handle a specific action as part of deprovisioning. For example, an action could sign a user out, update a user profile, delete tokens, or get a list of user groups. In general, each action corresponds to a specific Google Workspace API endpoint (e.g. users.get in the Google Admin SKD API).
- Notification
 - Flows that sending various notifications to different systems based on the results of actions performed during deprovisioning. This may include sending notifications like sending a message to a Slack thread or adding a comment to a GitLab issue.
- Log
 - Flows that log various offboardings and events as part of deprovisioning. Log flows may log the results of a specific action, an new offboarding, updates to an offboarding, etc.

Connectors

Google Deprovisioner uses a combination of pre-built 3rd party connectors and custom connectors built with connector builder. Connectors can be found in the connection page of Workflows. The following connectors are used

- Google Workspace
- Google Workspace Enterprise License Manager
- Google Data Transfer
- Gmail
- Custom Slack

Tables

OWF does not provide native database support, so tables are used at this time to store logs. Tables are updated by the log flows.

The following tables are used

- Offboarding
 - contains data about each offboarding
- Events
 - contains data about a specific action that was performed as part of an offboarding

Offboarding and Event Objects

By Workflows standards, a large amount of data needs to be passed between the various flows that consistute Google Deprovisioner. To facilitate the exchange of information across all the

various different flows, the deprovisioner uses two standardized JSON objects.

Offboarding Object

The offboarding object contains all information related to a given offboarding. This includes user (including manager), actor, and status information. The offboarding object is passed to each action, notification, and log flow during the deprovisioning process.

- id: a unique identifier used to identify each offboarding. Derived from the flow's execution id
- user: contains relevant profile data about the user being offboarded
 - user also contains profile data about the user's manager
- actor: contains information about the user that triggered the offboarding
- status: the current status or state of the offboarding

An example of the offboarding object can be found inside of each orchestrator flow.

Event Object

The event object contains all information related to a given action. This includes

- id: a unique identifier used to identify each action event. Derived from the flow's execution id
- name: a display name for each action
- timestamp: timestamp of when the action was triggered
- response: results of the action
 - statusCode: status code returned from the API call used to perform an action
 - message: message returned from the API call used to perform an action
 - context: additional information about the action results. This may include execution history or error details depending on the results.

Each Action flow generates an event object every time its run.

How it works

Below is a high level overview of how Google Deprovisioner operates. For a more detailed overview read the comments and notes provided in each flow.

Initial Offboarding

- Trigger - /google-offboard <user email>
 - User triggers the Slack slash command and includes email of user to offboard
 - Verifies an email is present and gather initial user and actor data. required by the orchestrator flow.
 - Start orchestrator flow.
- Orchestrator
 - Construct offboarding object from initially provided data
 - Log - new offboarding with initiated
 - Notification - new offboarding
 - Action - get user's profile data

- Action - get manager's profile data
- Verify whether user's profile data present
 - If Present
 - Action - perform offboarding actions
 - Log - result of each action to events
 - Notification - Send action results
 - Notification - completed offboarding
 - Notification - update manager about offboarding
 - Log - update offboarding with Completed
 - If Missing
 - Notification - cancelled offboarding
 - Log - update offboarding with Cancelled

Post 90-Day Offboarding

- Trigger - Scheduled
 - Run daily and get list of all past offboardings. If number of days since offboard is
 - 60 days
 - Notification - Manager reminder
 - 83 days
 - Notification - Manager reminder
 - >= 90 days
 - Notification - Manager reminder
 - gather offboarding data and trigger orchestrator flow
- Orchestrator
 - Construct offboarding object from offboarding data
 - Notification - new post-90 offboarding
 - Action - get user's profile data
 - Action - get manager's profile data
 - Verify whether user's profile data present
 - If Present
 - Action - perform offboarding actions
 - Log - result of each action to events
 - Notification - Send action results
 - Notification - post-90 completed offboarding
 - Log - update post-90 offboarding with Completed
 - Missing
 - Notification - cancelled post-90 offboarding
 - Log - update post-90 offboarding with Cancelled

SECTION: security/corporate/systems/sentinelone/_index.md

Overview

We use SentinelOne for endpoint (team member laptop) detection and response (EDR) at GitLab.

All macOS, Windows and Linux devices used by GitLab Team Members for the purposes of fulfilling the responsibilities of their role as a GitLab Team Member are required have the SentinelOne

EDR agent installed and functioning.

The use of a Windows endpoint requires a specific business reason and an approved exception as the use of a Windows endpoint is prohibited.

- sentinelone Slack Channel
- SentinelOne Setup Guide
- SentinelOne Troubleshooting Guide

DRI

This platform has a shared responsibility model.

- Corporate Security (Device Trust Engineering): Configuration and deployment of agents (via Jamf)
- Security Operations (Security Incident Response Team): Manage console alerts for incidents and investigations
- Security Compliance: Auditing components and configuration
- Legal Privacy: Audit of incidents and appropriate administrator access to personal data

Background Context

GitLab is a growing global company and as such we are required to meet compliance requirements in different countries, with one of those requirements being endpoint protection. An EDR solution meets that requirement. A number of customers also ask us if we are protecting their data by us protecting the systems that could potentially access their data, and ask specifically about EDR and Antivirus. EDR meets this requirement and addresses customer concerns.

Equally important is the security and privacy of our team members and their laptops. As GitLab continues to grow, we are increasing the likelihood that we will be targeted by malicious attackers. We want to ensure we have a way to protect team members and their laptops whether they are at home, on the road, or at the coffee shop down the street.

With an increased number of high profile breaches in the media today it is important we carefully consider the risk and external threats GitLab faces. Due to our service and the companies that use us, it is credible that we are or will soon become a high profile target for adversaries looking to exploit GitLab or a GitLab customer.

Vendor Decision

- EDR Proof of Concept Repository

How It Works

Endpoint detection and response (EDR) technology helps further strengthen endpoint security protecting not only GitLab, but each of our Team Members as well. EDR is a collection of endpoint security technologies that together record and store endpoint-system-level behaviors, use various data analytics techniques to detect suspicious system behavior, provide contextual information, block malicious activity, and provide remediation paths to restore affected

systems.

The first line of defense is stopping (or if configured, alerting) common threats on the laptop. As for additional or more "layered" lines of defense, certain types of activity (mainly involving processes on the laptop) are recorded and uploaded to a private instance for further processing and investigation. Much more sophisticated processing can take place for more thorough threat detection. As a result, the agent on the laptop is typically 40-60mb in size, as opposed to the older traditional antivirus solutions which can be well over a gigabyte. This results in less resources being used on the laptop including low CPU and low memory usage.

- Wikipedia - Endpoint Security
- Wikipedia - Endpoint Detection and Response
- Wikipedia - Data Loss Prevention Software
- Wikipedia - Network Detection and Response

Antivirus vs EDR

If you remember the old days of installing antivirus and anti-malware scanners on your computer, this is similar next-generation of scanning technology with a wider reach to improve our security posture.

Antivirus can be seen as one part of the EDR complete security technology solution. Traditionally, antivirus software is a single program which serves basic purposes like scanning, detecting and removing viruses and different types of malware. For EDR, real time status, remediation paths, endpoint firewalls, and system behavioral analytics come together to create a comprehensive technology stack to protect against modern day digital attacks.

MDM vs EDR

The EDR technology is in addition to Jamf MDM. While Jamf does offer security features and protections, an EDR solution combined with Jamf will offer great ability to detect and stop significant threats and advanced digital attacks on GitLab team member laptops.

FAQs

Monitoring Exemption

Do not attempt to modify or remove SentinelOne components, unless instructed to do so by Corporate Security for troubleshooting purposes.

If SentinelOne is causing issues with your ability to fulfill the responsibilities of your role, you may request an exception to be granted using the Information Security Policy Exception Management Process.

Exceptions will be granted based on the goal of least needs, attempting to maximize effectiveness, efficiency, and security risk mitigation of SentinelOne while minimizing the negative impact on the team member. For this reason, rarely will an exception be granted to completely disable SentinelOne. Rather, an exception will be granted for specific directory paths, specific applications, or specific collections of data that SentinelOne may be negatively impacting through quarantine or system performance degradation.

Local Network Monitoring

> Will it monitor my local network?

No. The EDR solution only monitors the GitLab laptop, like an antivirus solution would (only much more efficiently).

Home Network Privacy

> What options does a team member have to protect their home network privacy?

If you wish to add further privacy and security to your home network, you can further isolate your work machine by creating a separate network for it.

While we cannot provide you with any direct support for this type of network setup, see the wireless networks and network isolation guides. examples that might help to get you started.

Third Party Vendor

> Why are we using a third party EDR system?

We do not want to write our own EDR solution, nor do we want to assemble a series of partial solutions into some type of chain to attempt to meet compliance requirements. Not only does a third party EDR solution allow us to meet requirements in a thorough and more economic way, it will show auditors, regulators, and customers that we are taking security seriously and implementing it consistently.

SaaS vs Self Hosted

> Can we self-host our own solution to have more control?

We could, although this would require a substantial amount of effort to be able to have a self-managed solution that would address both compliance requirements and customer concerns. The chosen EDR technology is a single-tenant-hosted third party EDR solution to help ensure GitLab and Team Member data privacy.

Operating System Partitions

> I dual boot two different operating systems, do they both need an agent?

Yes. Each operating system on a host computer that is used to access GitLab computing resources, infrastructure, or environments, will need have an EDR agent installed.

Virtual Machines

> I have several virtual hosts/machines on my laptop, do they all need agents?

Yes. Each operating system on a computer, to include the host operating system, and all guest operation systems, that are used to access GitLab computing resources, infrastructure, or environments, will need have an EDR agent installed.

Antivirus Alternative

> I run my own antivirus solution on my work laptop, isn't that enough?

While this effort is certainly appreciated, we need a way to be able to audit laptops to meet compliance requirements. We would still encourage you to run any such antivirus product up until the EDR solution has been rolled out to your laptop. After which time, we strongly encourage you to uninstall the antivirus solution previously installed. Multiple instances of these technologies, especially from different vendors, have the tendency to cause issues on an endpoint which may negatively impact system performance and your productivity.

Administrative Safeguards

> What safeguards are in place to ensure that owners of this process can prevent abuse?

While such a possibility exists, we feel that the risk of something like this happening is much, much smaller than some of the risks that an endpoint management solution is made to address. Risks like:

- Laptop is accidentally infected with malware and team-member is unaware of the infection.
- Laptop is accidentally infected with malware or was the target of an attack.
- The laptop is compromised and the GitLab team members access and identity are used to further attack GitLab and our customers.
- A new security exploit is discovered, and unpatched applications start getting exploited in the wild.
- GitLab experiencing a targeted attack on our endpoints, which could lead to a compromise of the GitLab Production environment resulting in a customer data breach.

That being said, the EDR solution provides the ability to audit any processes and actions that have been conducted by an Administrator or the technology itself. The solution has role-based access controls implemented to limit access to certain features, and all of the use of such features is auditable.

SECTION: security/corporate/systems/sentinelone/setup/_index.md

macOS

> No Action Required: SentinelOne is automatically installed by Jamf MDM and you do not need to perform any installation tasks.

Linux

Pre-Requisites

If you are using Advanced Intrusion Detection Environment (AIDE) to monitor file integrity and detect intrusions, you will need to create an exclusion in AIDE. When both AIDE and the SentinelOne Agent are running together, AIDE is not able to update its database correctly. AIDE tries to scan a SentinelOne Directory and cannot scan it.

To create an exclusion for SentinelOne, edit `/etc/aide.conf` and add `!/opt/sentinelone/mount` to the end to ignore the SentinelOne Agent mount directory:

```
shell
echo '!/opt/sentinelone/mount' | sudo tee -a /etc/aide.conf
```

Installation

> These steps are only needed for Linux laptops. SentinelOne is already automatically installed on macOS by Jamf MDM.

1. Make sure you are using an approved Linux distribution.

1. Download the configuration file and the installer (DEB/RPM).

1. In the following examples, we'll assume you put those in the following locations and that you renamed the package name to be `sentinelagent`.

```
> Configuration file: ~/.config/sentinelone/config.cfg
> DEB package: ~/Downloads/sentinelagent.deb
> RPM package: ~/Downloads/sentinelagent.rpm
```

1. Get your laptop's serial number.

```
shell
sudo dmidecode -s system-serial-number
```

1. Edit `config.cfg` and update `S1AGENTCUSTOMERID`.

- Replace `tanuki` with the username portion (the part before the `@` sign or the local-part per RFC2822) of your GitLab email address.
- Replace `ABCD1234` with your laptop's serial number.
- Verify that the edited variable is formatted correctly with a hyphen separating the GitLab email and serial number. For example, `S1AGENTCUSTOMERID=jdoe@gitlab.com-ABCD1234`.

1. Install the package (distro commands may vary).

```
shell
Fedora 37+ and RPM distributions
export S1AGENTINSTALLCONFIGPATH=~/.config/sentinelone/config.cfg
sudo -E rpm -i --nodigest ~/Downloads/sentinelagent.rpm
```

```
shell
Ubuntu 22.04+ and Debian/Ubuntu distributions
export S1AGENTINSTALLCONFIGPATH=~/.config/sentinelone/config.cfg
sudo -E apt install ~/Downloads/sentinelagent.deb
```

- ```
plaintext
Setting registration token...
Registration token successfully set
Setting management device type...
Device type successfully set
Setting customer ID...
Customer ID successfully set
Starting agent...
Agent is running
```

- ```
shell
systemctl status sentinelone
```

1. If you see the previous step outputs, you're all set!

— — —

```
## SECTION: security/corporate/systems/sentinelone/troubleshooting/ index.md
```

If you are experiencing issues with SentinelOne, please try the following troubleshooting steps.

Linux Installation Error

> I do not see the outputs on the Linux installation steps. What should I do?

The SentinelAgent was not installed correctly. Most likely this is due to a configuration file error or commands did not run successfully.

1. Verify that you are using a supported Linux distribution.

1. Uninstall the package.

shell

Fedora 37+ and RPM distributions

```
sudo dnf remove sentinelagent
```

shell

Ubuntu 22.04+ and Debian/Ubuntu distributions

```
sudo apt purge sentinelagent
```

1. Install the package again (distro commands may vary).

shell

Fedora 37+ and RPM distributions

```
export S1AGENTINSTALLCONFIGPATH=~/.config/sentinelone/config.cfg
```

```
sudo -E rpm -i --nodigest ~/Downloads/sentinelagent.rpm
```

shell

Ubuntu 22.04+ and Debian/Ubuntu distributions

```
export S1AGENTINSTALLCONFIGPATH=~/.config/sentinelone/config.cfg
```

```
sudo -E apt install ~/Downloads/sentinelagent.deb
```

1. Verify that the agent is running in the terminal output.

plaintext

```
Setting registration token...
```

```
Registration token successfully set
```

```
Setting management device type...
```

```
Device type successfully set
```

```
Setting customer ID...
```

```
Customer ID successfully set
```

```
Starting agent...
```

```
Agent is running
```

1. Wait five minutes, and then verify connectivity:

```
shell
sudo sentinelctl management status
```

1. You should see Connectivity: On and a valid SentinelOne URL. If this is not your result, reach out for assistance in the sentinelone channel.

CPU Usage

- > How do I calculate how much CPU is being used by SentinelOne on MacOS?

If you are using Activity Monitor to monitor CPU usage, be aware that it is calculated as the percent utilized for a single CPU thread. For example, an M1 Max laptop has 10 threads, so total CPU capacity is 1000%. To view the number of threads available on your system, open Terminal and run:

```
shell
sysctl -n hw.ncpu
```

Expect SentinelOne to use less than 10% of total CPU power (for example, displayed as less than 100% in Activity Monitor on a M1 Max).

Usage Metrics

- > How do I collect metrics for support on MacOS due to an issue with high CPU or RAM?

1. Open your Terminal or iTerm and run the following command.

```
shell
sudo sentinelctl profiler start 30
```

2. Reproduce the scenario that caused the issue, or run it for a few minutes. It will run in the background. You will not see output.

3. Collect the metric log:

```
shell
sudo sentinelctl profiler stop
```

4. You should see a notification in the top right on your screen indicating a report is being generated.

5. After some time a report will finish and you will see another notification from SentinelOne that a log file was completed. Click the notification and you may be required to allow the SentinelAgent access to files on your Desktop folder. The file will then download to your Desktop folder.

6. Analyze the data that the report generates, and if needed share the file in the sentinelone Slack channel and we will get a ticket open with SentinelOne support.

Offline Agent

The most common reason for a SentinelOne agent to appear offline is a local firewall prohibiting outbound connections to .sentinelone.net. HTTPS (port 443) needs to be allowed outbound to that domain.

For macOS, the most common local firewalls are Little Snitch, pfSense or Lulu.

Linux users may have pfSense or iptables blocking access.

macOS Offline Agent

1. Open your Terminal or iTerm and run the following command.

```
shell
sudo sentinelctl status
```

1. Look for the status of Connected: (should be "on").

1. Resolve the local firewall block and it should re-connect.

Linux Offline Agent

1. Open your Terminal and run the following command.

```
shell
sudo sentinelctl management status
```

1. You should see Connectivity: On and a valid SentinelOne URL.

1. If this is not your result, reach out for assistance in the sentinelone channel.

SECTION: security/corporate/systems/slack/_index.md

This is a placeholder page. Please see the links below for any child pages that exist.

SECTION: security/corporate/systems/slack/apps/_index.md

GitLab has chosen to restrict the ability to install apps, and we have a process to approve or restrict certain apps for our workspace. In order to integrate a new app with existing applications in our tech stack such as Google, Slack or Zoom you need to create a Third Party Risk Management issue.

Once the TPRM issue has been approved by all parties, please request approval to add the app to Slack following the steps below:

1. Open a CorpSec Slack app request in our tracker to request integration.
1. Within Slack, scroll to the "Apps" section. Click the "+ Add apps".
1. Search for the app that you would like to add, and click "Add".
1. A web browser will open. Please click on Add to Slack.
1. Add a custom message with more context about your request and also link the TPRM approval issue+CorpSec issue.
1. Click Submit. You'll receive a direct message from Slackbot when your request has been reviewed by the team.

SECTION: security/corporate/systems/slack/channels/_index.md

This is a placeholder page. Please see the links below for any child pages that exist.

SECTION: security/corporate/systems/slack/external-users/_index.md

This is a placeholder page. Please see the links below for any child pages that exist.

SECTION: security/corporate/systems/slack/groups/_index.md

Temporary Links

- Slack Group Membership Reports

SECTION: security/corporate/systems/slack/setup/_index.md

This is a placeholder page. Please see the links below for any child pages that exist.

SECTION: security/corporate/systems/slack/webhooks/_index.md

This is a placeholder page. Please see the links below for any child pages that exist.

SECTION: security/corporate/systems/teleport/_index.md

Teleport is an Access Management Platform allowing users to gain read or write access to various resources.

The infrastructure runbook docs contain a variety of detailed topics on using and interacting with teleport.

Until CorpSec is staffed to provide technical ownership, Foundations providing maintenance of the underlying infrastructure. All non-maintenance related feature requests and improvements are managed by CorpSec.

For help with:

- Business ownership (contract, compliance, roadmap)- contact CorpSec in security-corpsec.
- Access - open an access request.
- Individual access approvals - Follow the runbook for connecting to a database console or rails console.
- Onboarding new resources - open a new issue for CorpSec to triage and schedule.
- Feature requests or improvements - open a new issue for CorpSec to triage and schedule.
- Maintenance work - open an issue in production-engineering or ask questions in gfoundations.

SECTION: security/corporate/systems/trainingctl/_index.md

This is a placeholder page. Please see the links below for any child pages that exist.

SECTION: security/corporate/systems/vpn/_index.md

Overview

NordLayer is our supported VPN (Virtual Private Network) platform for GitLab Team Members.

The use of NordLayer is optional, however it is recommended when working on guest networks or public Wi-Fi. In other words, you should connect to the VPN to secure your laptop's traffic anytime that you're not at home. That could be at a co-working location, an airport, a coffee shop or on a guest network at a customers office.

Some members may use a different VPN solution for these scenarios and we would recommend that they migrate to NordLayer when it's convenient.

- Wireless Networks Security Standards
- Setup Guide

- Troubleshooting Guide
- Vendor Docs

Background Context

When we first approached the idea of a simple VPN for those times that our employees work away from home (coffee shops, trains, planes, etc.), we found a lot of options out there. While many VPN options exist, some are much more than we need, and some far less.

When it comes to our decision on NordLayer, we had a number of things to look at:

- Is it secure?
- Is it easy to administer?
- Does it support multiple operating systems?
- Are admin actions logged?

We tested many options and while a number fit a few of these, NordLayer fit the most with security being our most important criteria.

System Owner and Support

- DRI: @cshankgitlab
- security-corpsec Slack channel

SECTION: security/corporate/systems/vpn/setup/_index.md

Access Request

Please request access to NordLayer through the Okta portal using Lumos. Follow these steps to submit your request for NordLayer:

- Log in to the Okta portal
- Search for and select the "Lumos" application
- Select "NordLayer" from the available applications
- Choose the specific permission that matches your needs, either "Public WiFi" or "Dedicated Gateway"
- Lumos will automatically initiate the approval workflow and handle provisioning upon approval

The system will guide you through any additional information required to complete your request. You'll receive notifications about the status of your request through email.

NordLayer Installation

After your access request is provisioned, you will receive an email from nordlayer@nordlayer.com with a link to download the application.

- MacOS:

1. Enter gitlab as the organization.
2. Authenticate with Okta.
3. When prompted, select to Allow an upcoming prompt.

- Linux:

1. Follow <https://help.nordlayer.com/docs/installing-on-linux> to download the application and add your user to the user group.
1. Log out and log back in.
1. In your terminal, run nordlayer login.
1. Enter gitlab as the organization.
1. Open the URL you get in your browser and authenticate with Okta.

- Android:

1. Go to Google Play and search for NordLayer. Tap Install and download the app.
1. Enter gitlab as the organization ID and tap continue.
1. Tap Log in with Okta and follow the prompts.
1. Tap Allow access to allow NordLayer access to your VPN settings.
1. Tap Ok NordLayer to set up a VPN connection.
1. Tap Allow access to allow NordLayer access to your location (for recommended servers).
1. Tap Allow access to access VPN settings and you're all set!

- iOS:

1. Go to the App Store and search for NordLayer and tap Get to download the app.
1. Enter gitlab as the organization ID and tap continue.
1. Tap Log in with Okta and follow the prompts.
1. Tap Allow access to allow NordLayer access to your VPN settings.
1. Tap Allow to allow NordLayer to add VPN configurations.
1. Tap Allow access to allow NordLayer access to your location (for recommended servers) and you're all set!

Configuration

We recommend you set NordLayer to auto-connect when using untrusted WiFi (networks with no password or weak encryption):

- MacOS:

1. Go to NordLayer preferences and select the Auto-Connect tab.
1. Select When using untrusted Wi-Fi (or When app launches and add your home network as trusted).

- Linux:

1. If NordLayer is running, in your terminal run nordlayer settings set.
1. Select Auto-connect, and then When using untrusted Wi-Fi.

To learn more, see NordLayer application usage on Linux.

Add Your Home Network to the Trusted List (MacOS only)

> When you add a trusted network to the list, your laptop will keep NordLayer disconnected while at home. This allows you to use the internet at full speed, and will automatically connect while you're on any untrusted network away from home.

>

> Please keep in mind that when you connect to a trusted network, you are no longer protected through a VPN. Do not do this if you're at a hotel, on a guest network, or anywhere outside of your home network (unless needed to get through initial wi-fi portal terms and conditions captive portal screen).

1. Go to NordLayer preferences and select the Auto-Connect tab.

1. Verify the Network listed under Current Network is your home network and select Trust.

NordLayer for System Administration

NordLayer is also used for system administration purposes, providing secure access to internal systems and resources via dedicated IP addresses that are limited to certain GitLab Team Members. If you require access to these dedicated IP addresses, please request the access via Lumos and select Permission:Dedicated Gateway.

Here are some key points about using NordLayer for system administration:

1. Access Control: NordLayer allows for granular access control, ensuring that only system administrators can access specific resources.

2. Multi-Factor Authentication: For enhanced security, NordLayer is implemented with Okta Device Trust Authentication policies.

3. Secure Remote Access: System administrators can securely access internal systems from any location, enabling efficient remote work and incident response.

4. Network Segmentation: NordLayer supports network segmentation, allowing administrators to isolate sensitive systems and limit potential security risks.

5. Encrypted Communication: All traffic between the administrator's device and internal systems is encrypted, protecting sensitive data in transit.

6. Centralized Management: The NordLayer admin panel provides a centralized interface for managing user access, monitoring connections, and configuring security policies.

When using NordLayer for system administration, always follow GitLab's security best practices and ensure that you have the appropriate permissions before accessing any systems or data.

Dedicated IP's are listed below for reference by Security teams or incident response.

| Location | IP |

| --- | --- |

| GitLab - Los Angeles | 216.74.107.115 |

| GitLab - New York | 146.70.186.59 |

GitLab - Atlanta	205.234.251.167
GitLab - Belgium	146.70.55.7
GitLab - Hungary	217.138.192.12
GitLab - Japan	146.70.138.86
GitLab - Australia	88.216.59.30

SECTION: security/corporate/systems/vpn/troubleshooting/_index.md

Overview

- Vendor Docs

FAQs

Linux VPN Protocol

In case you are experiencing any issues with your NordLayer connection, the first course of action would be to change the VPN protocol. You can do so by running the following command `nordlayer settings set` and selecting VPN protocol.

Windows 10 Sufficient Privileges

Verify that you have sufficient privileges to start system services

If you are getting an error 'Verify that you have sufficient privileges to start system services' on Windows 10 while installing the NordLayer application - please make sure that you are using an up-to date Windows 10 version.

Lin